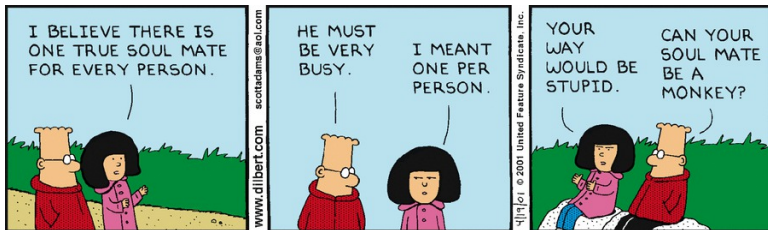


Logic and Sets

6

Predicate logic



Today

Syntax of predicate logic

- ▶ atomic formulas, built from:
 - ▶ predicate symbols (like *Person*(_) or *Soulmate*(_,_))
 - ▶ constants (we will e.g. use *s* for Socrates)
 - ▶ variables (like *x* and *y*)
- ▶ connectives of propositional logic
- ▶ universal quantification $\forall x \phi$
- ▶ existential quantification $\exists x \phi$

Semantics of predicate logic

- ▶ models
- ▶ typical semantic equivalences and non-equivalences

Syntax of predicate logic



The need for a richer language

- ▷ All human beings are mortal.
- ▷ Socrates is a human being.
- ▷ Therefore, Socrates is mortal.

Oversimplistic translation key

- p all human beings are mortal
- q Socrates is a human being
- r Socrates is mortal

$p, q \models r$ is a semantic entailment that does not hold !

The semantic entailment must reflect:

- ▶ Being mortal is a property of all elements in the set of human beings.
- ▶ Socrates is an element of this set.

Argument formalization in predicate logic

- ▷ All human beings are mortal.
- ▷ Socrates is a human being.
- ▷ Therefore, Socrates is mortal.

Translation key

predicate $H(x)$: x is a human being

predicate $M(x)$: x is mortal

constant s : Socrates

Formalization:

$$\forall x (H(x) \rightarrow M(x)), H(s) \models M(s)$$

This semantic entailment will turn out to hold.

Predicates, constants, and variables

In *predicate logic*, propositional variables p, q are replaced by **atomic formulas**, with *predicates*, *constants* and *variables*.

Constants are concrete objects, *variables* range over all objects, *predicates* express properties of and relations between objects.

Example: C is a **1-ary (unary)** *predicate symbol*, expressing that its argument is clever.

J is represented by the *constant* j .

$C(j)$ expresses that J is clever.

Example: K is a **2-ary (binary)** *predicate symbol*, expressing that its first argument knows its second argument.

A and B are represented by the *constants* a and b .

$K(a, b)$ expresses that A knows B .

Example: L is a **3-ary (ternary)** *predicate symbol*, expressing that its three arguments are situated on a line.

Logical connectives

Atomic formulas are now the smallest building blocks.

The connectives $\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$ are used to construct formulas.

Examples

► $C(j) \wedge K(a, b)$

► $\neg K(a, a)$

► $K(a, b) \wedge \neg K(b, a)$

Possible translation key

j : Jane

a : Anton

b : Betty

$C(x)$: x is clever

$K(x, y)$: x knows y

Quantifiers

Instead of *Jane is clever*, we now want to express:

somebody is clever.

A quantifier is needed, in this case $\exists$, the **existential quantifier**:

$$\exists x C(x)$$

For *everyone is clever* one uses $\forall$, the **universal quantifier**:

$$\forall x C(x)$$

Examples

► $\neg \forall x C(x)$

$$\exists x \neg C(x)$$

► $\forall x (K(x, x) \rightarrow C(x))$

► $\forall x \neg K(x, x)$

$$\neg \exists x K(x, x)$$

Predicate logic: Syntax

- ▶ **predicate**, **constant**, and **variable** symbols, like:
 - ▶ C , K are predicate symbols (with arities 1, 2)
 - ▶ a , b , j are constants
 - ▶ x , y , z are variables
- ▶ **atomic formulas**, consisting of a predicate with *constants* and *variables* as arguments, are the building blocks, like:

$$C(j), \quad K(a, b), \quad C(x), \quad K(x, y)$$

- ▶ **connectives** $\neg \vee \wedge \rightarrow \leftrightarrow$ to compose formulas
- ▶ **quantifiers** $\forall$ ('for **A**ll') $\exists$ ('there **E**xists')
- quantifications $\forall x \exists x$ are 1-ary connectives (just as $\neg$)

Priority rules

The priority level for **quantifications** is the same as for $\neg$:

$$\frac{\neg \quad \forall x \quad \exists x}{\wedge \quad \vee} \quad \frac{\quad}{\rightarrow \quad \leftrightarrow}$$

Examples:

- ▶ $\forall x (C(x)) \wedge K(a, j)$ can be written as $\forall x C(x) \wedge K(a, j)$
- ▶ $\forall x (C(x) \wedge K(a, x))$ does need the outer pair of brackets
- ▶ $\neg(\forall x (C(x)))$ can be written as $\neg \forall x C(x)$

Bound and free variables

An occurrence of a variable x in a formula is **bound** if it is *in the scope* of a $\forall x$ or $\exists x$. Else it is **free**.

Examples:

- ▶ In $\exists x (C(x) \wedge D(x))$, *both* occurrences of x behind the quantifier are *bound*.
- ▶ In $\exists x C(x) \wedge D(x)$, the occurrence of x in $C()$ is *bound* while the occurrence of x in $D()$ is *free*.
- ▶ In $\exists x C(x) \wedge \exists x D(x)$ the occurrences of x in $C()$ and in $D()$ are *bound* by the first and second $\exists x$, respectively.

Question

Say for each occurrence of x in $\forall x (\exists x C(x) \wedge D(x))$ whether it is bound, and if so, by which quantifier.

Semantics of predicate logic

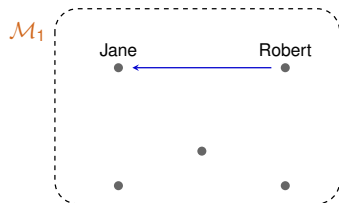


All fish are mortal.
I am mortal.
Therefore, I am a fish???

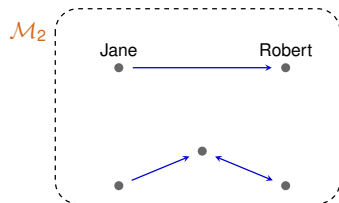
Models for predicate logic

$L(r, j)$ expresses that Robert loves Jane.

This relation is depicted below by blue arrows.



$\mathcal{M}_1 \models L(r, j)$



$\mathcal{M}_2 \not\models L(r, j)$

In model $\mathcal{M}_1$ (at the left) this holds.

But in model $\mathcal{M}_2$ (at the right) it does not.

Predicate logic: Models

The *meaning* (or *truth value*) of a formula in predicate logic depends on the underlying **model** $\mathcal{M}$, consisting of:

- ▶ a non-empty **set** A of elements (**Jane**, **Robert**)
- ▶ an **interpretation**, with regard to elements in A , of the **constants**, **free variables**, and **predicate symbols**.

$\forall x \phi$ holds in model $\mathcal{M}$ if ϕ holds for *every* element x in A :

For *each* $e \in A$, the formula that is obtained by substituting e for the **free** occurrences of x in ϕ holds in $\mathcal{M}$.

$\exists x \phi$ holds in model $\mathcal{M}$ if ϕ holds for *some* element x in A :

For *some* $e \in A$, the formula that is obtained by substituting e for the **free** occurrences of x in ϕ holds in $\mathcal{M}$.

An example of a model

For interpreting the formulas

Jane is clever and Robert loves her $C(j) \wedge L(r, j)$

not everyone is clever $\neg \forall x C(x)$

we use a *model* $\mathcal{M}$ consisting of:

- ▶ a **set** of elements; quantifiers range over this set
- ▶ an **interpretation** of the constants, free variables, and predicate symbols

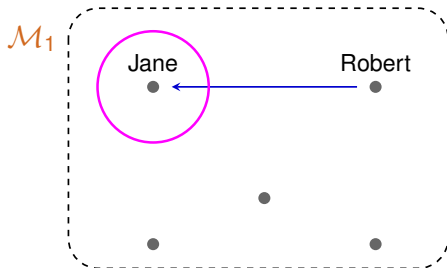
Example *model* $\mathcal{M}$

- ▶ A = all human beings (set)
- ▶ $j^{\mathcal{M}} = \text{Jane}$, $r^{\mathcal{M}} = \text{Robert}$ (elements)
- ▶ $C^{\mathcal{M}} = \{e \in A \mid e \text{ is clever}\}$ (subset)
- ▶ $L^{\mathcal{M}} = \{\langle e, f \rangle \mid e \in A, f \in A, e \text{ loves } f\}$ (relation)

Semantics of predicate logic: Model 1

Here we are in a hypothetical world (model $\mathcal{M}_1$) where:

- ▶ $j^{\mathcal{M}_1} = \text{Jane}$, $r^{\mathcal{M}_1} = \text{Robert}$
- ▶ $L^{\mathcal{M}_1} = \{\langle \text{Robert}, \text{Jane} \rangle\}$
 - ▶ Robert loves Jane (blue arrow)
 - ▶ nobody else loves anybody
- ▶ $C^{\mathcal{M}_1} = \{\text{Jane}\}$
 - ▶ Jane is clever (pink circle)
 - ▶ nobody else is clever



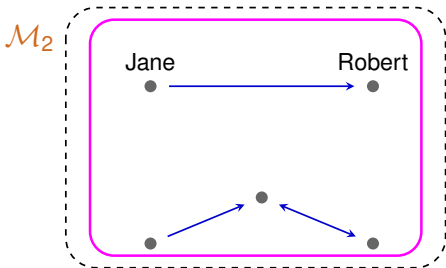
For $\mathcal{M}_1$ we find:

$$\begin{array}{lll} \mathcal{M}_1 \models C(j) & \mathcal{M}_1 \models L(r, j) & \mathcal{M}_1 \models C(j) \wedge L(r, j) \\ \mathcal{M}_1 \not\models C(r) & \mathcal{M}_1 \not\models \forall x C(x) & \mathcal{M}_1 \models \neg \forall x C(x) \end{array}$$

Notation (for truth in a model $\mathcal{M}$)

For a formula ϕ , we denote by $\mathcal{M} \models \phi$ that ϕ holds in $\mathcal{M}$.

Semantics of predicate logic: Model 2



Here we are in a hypothetical world (model $\mathcal{M}_2$) where:

- ▶ $j^{\mathcal{M}_2} = \text{Jane}$, $r^{\mathcal{M}_2} = \text{Robert}$
- ▶ $L^{\mathcal{M}_2} = \{\langle \text{Jane}, \text{Robert} \rangle, \dots\}$
 - ▶ Jane loves Robert (blue arrow)
 - ▶ ... loves ... (other blue arrows)
- ▶ $C^{\mathcal{M}_2} = \{x \mid x \text{ is human being}\}$
 - ▶ everyone is clever (indicated by the pink area)

For $\mathcal{M}_2$ we find:

$$\mathcal{M}_2 \models C(j)$$

$$\mathcal{M}_2 \not\models L(r, j)$$

$$\mathcal{M}_2 \not\models C(j) \wedge L(r, j)$$

$$\mathcal{M}_2 \models L(j, r)$$

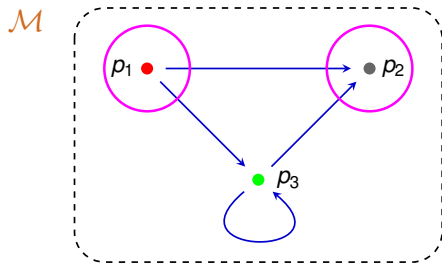
$$\mathcal{M}_2 \models C(j) \wedge L(j, r)$$

$$\mathcal{M}_2 \models C(r)$$

$$\mathcal{M}_2 \models \forall x C(x)$$

$$\mathcal{M}_2 \models \exists x \exists y (L(x, y) \wedge L(y, x))$$

Interpreting formulas *without* quantifiers (example)

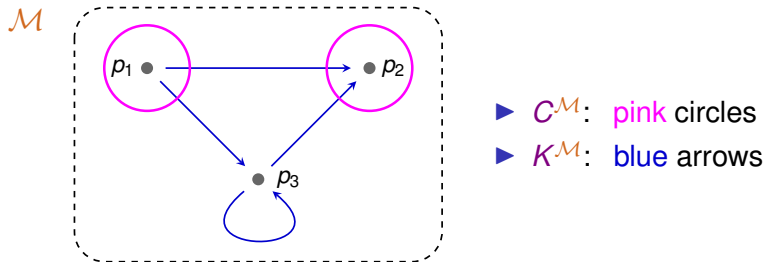


- ▶ $\mathcal{M} \models K(a, b)$
- ▶ $\mathcal{M} \not\models K(b, a)$
- ▶ $\mathcal{M} \models \neg K(b, a)$
- ▶ $\mathcal{M} \models K(b, b)$
- ▶ $\mathcal{M} \models C(a)$
- ▶ $\mathcal{M} \not\models C(b)$
- ▶ $\mathcal{M} \models \neg C(b)$

Model $\mathcal{M}$ with unary predicate C , binary predicate K

- ▶ $A = \{p_1, p_2, p_3\}$
- ▶ interpretations of constants a, b and predicates C, K in $\mathcal{M}$:
 - ▶ $a^{\mathcal{M}}$: red point p_1
 - ▶ $b^{\mathcal{M}}$: green point p_3
 - ▶ $C^{\mathcal{M}}$: pink circles
 - ▶ $K^{\mathcal{M}}$: blue arrows

Interpreting formulas *with* quantifiers (example)



► $\mathcal{M} \not\models \forall x C(x)$

$\mathcal{M} \models \neg \forall x C(x)$

► $\mathcal{M} \models \exists x C(x)$

$\mathcal{M} \models \exists x \neg C(x)$

► $\mathcal{M} \not\models \forall x (C(x) \rightarrow \exists y K(x, y))$

Question: Which of the following formulas are true in $\mathcal{M}$?

► $\mathcal{M} \models \forall x (C(x) \vee K(x, x))$

► $\mathcal{M} \not\models \exists x \forall y K(x, y)$

► $\mathcal{M} \models \forall x (\exists y K(x, y) \rightarrow \exists y (K(x, y) \wedge C(y)))$

Tautologies and contradictions

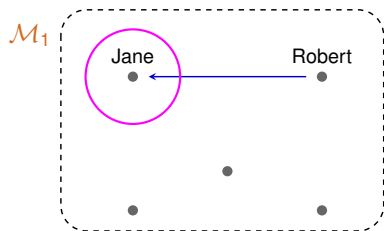
A *tautology* is true in all models

A *contradiction* is false in all models

A *contingent* formula is true in some model, false in another

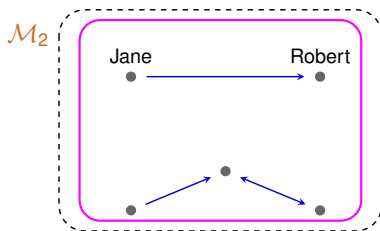
A *satisfiable* formula is true in some model

Two contingent formulas



$$\mathcal{M}_1 \models L(r, j) \wedge C(j)$$

$$\mathcal{M}_1 \not\models \forall x C(x)$$



$$\mathcal{M}_2 \not\models L(r, j) \wedge C(j)$$

$$\mathcal{M}_2 \models \forall x C(x)$$

So the formulas $L(r, j) \wedge C(x)$ and $\forall x C(x)$ are *contingent*.

Semantic entailment

Semantic entailment $\phi_1, \dots, \phi_n \models \psi$

- ▶ in *propositional* logic:

for all valuations v :

$$v(\phi_1) = \text{T} \text{ and } \dots \text{ and } v(\phi_n) = \text{T} \implies v(\psi) = \text{T}$$

- ▶ in *predicate* logic:

for all models $\mathcal{M}$:

$$\mathcal{M} \models \phi_1 \text{ and } \dots \text{ and } \mathcal{M} \models \phi_n \implies \mathcal{M} \models \psi$$

A **counterexample** for a semantic entailment $\phi_1, \dots, \phi_n \models \psi$ in predicate logic consists of a *model* $\mathcal{M}$ with

$$\mathcal{M} \models \phi_1, \dots, \mathcal{M} \models \phi_n \qquad \mathcal{M} \not\models \psi$$

Why is Socrates mortal, indeed ?

$$\forall x (H(x) \rightarrow M(x)), H(s) \models M(s) \quad (1)$$

This semantic entailment means that for *all* models $\mathcal{M}$:

$$\begin{aligned} \text{if } \mathcal{M} \models \forall x (H(x) \rightarrow M(x)) \text{ and } \mathcal{M} \models H(s), \\ \text{then also } \mathcal{M} \models M(s). \end{aligned} \quad (2)$$

Let $\mathcal{M}$ be a model with $\mathcal{M} \models \forall x (H(x) \rightarrow M(x))$ and $\mathcal{M} \models H(s)$.

- (i) $\mathcal{M} \models H(s)$ means that $s^{\mathcal{M}} \in H^{\mathcal{M}}$.
- (ii) $\mathcal{M} \models \forall x (H(x) \rightarrow M(x))$ means that $H^{\mathcal{M}} \subseteq M^{\mathcal{M}}$.
- (iii) By (i) and (ii) we conclude $s^{\mathcal{M}} \in M^{\mathcal{M}}$.
- (iv) This means $\mathcal{M} \models M(s)$.

So we have established (2) for all models, and thereby (1).

Is Socrates a fish ???

Question

Let $F(x)$ represent that x is a fish. Show that

$$\forall x (F(x) \rightarrow M(x)), M(s) \not\models F(s)$$

Answer: Let $A = \{p\}$, $F^{\mathcal{M}} = \emptyset$, $M^{\mathcal{M}} = \{p\}$ and $s^{\mathcal{M}} = p$.

(So there's Socrates, who is mortal, and there are no fish.)

$\forall x (F(x) \rightarrow M(x))$ and $M(s)$ hold in $\mathcal{M}$.

But $F(s)$ does not.



All fish are mortal.
I am mortal.
Therefore, I am a fish???

Semantic equivalence for predicate logic

Time for Tea

"Then you should say what you mean," the March Hare went on.

"I do," Alice hastily replied: "at least—at least I mean what I say—that's the same thing, you know."

"Not the same thing a bit!" said the Hatter. "You might just as well say that 'I see what I eat' is the same thing as 'I eat what I see!'"

"You might just as well say," added the March Hare, "that 'I like what I get' is the same thing as 'I get what I like!'"

"You might just as well say," added the Dormouse, who seemed to be talking in his sleep, "that 'I breathe when I sleep' is the same thing as 'I sleep when I breathe!'"



Nobody is perfect

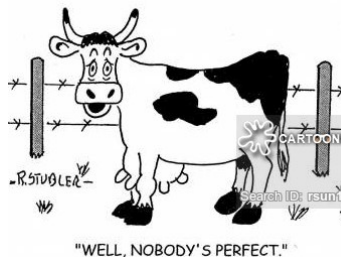
Translation key

$P(x)$: x is perfect

The following two sentences
have the same meaning.

nobody is perfect : $\neg \exists x P(x)$

everybody is imperfect : $\forall x \neg P(x)$



The two formulas are *semantically equivalent*:

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

Semantic equivalence

Semantic equivalence $\phi \equiv \psi$

- ▶ in *propositional* logic:

for all valuations v :

$$v(\phi) = \text{T} \iff v(\psi) = \text{T}$$

- ▶ in *predicate* logic:

for all models $\mathcal{M}$:

$$\mathcal{M} \models \phi \iff \mathcal{M} \models \psi$$

In predicate logic, just as in propositional logic,

$$\phi \equiv \psi \iff \phi \models \psi \wedge \psi \models \phi$$

$$\exists x (\phi \vee \psi) \equiv \exists x \phi \vee \exists x \psi$$

Question

Does the following semantic equivalence hold ?

$$\exists x (\phi \vee \psi) \equiv \exists x \phi \vee \exists x \psi$$

Answer: Yes ! “One of you is called Anne or Robert” is the same as “one of you is called Anne or one of you is called Robert”.

Question

Does the following semantic equivalence hold ?

$$\exists x (\phi \wedge \psi) \equiv \exists x \phi \wedge \exists x \psi$$

Answer: No ! “One of you is called Pete Johnson” is *not* the same as “one of you is called Pete and one of you is called Johnson”.

$$\exists x (\phi \vee \psi) \equiv \exists x \phi \vee \exists x \psi$$

To prove $\exists x (\phi \vee \psi) \equiv \exists x \phi \vee \exists x \psi$, we must establish that these two formulas hold in exactly the same models.

Let $\mathcal{M}$ be a model, with A as set of elements.

- (i) $\mathcal{M} \models \exists x (\phi \vee \psi)$
- (ii) *if and only if* for some $e \in A$, $\phi \vee \psi$ with e substituted for x holds in $\mathcal{M}$
- (iii) *if and only if* for some $e \in A$, ϕ or ψ with e substituted for x holds in $\mathcal{M}$
- (iv) *if and only if* $\mathcal{M} \models \exists x \phi$ or $\mathcal{M} \models \exists x \psi$
- (v) *if and only if* $\mathcal{M} \models \exists x \phi \vee \exists x \psi$.

α -conversion

$$\exists x C(x) \equiv \exists y C(y)$$

$$\forall x C(x) \equiv \forall y C(y)$$

Such renamings of *bound* variables are called α -conversion.

Hence,
$$\exists x C(x) \wedge \exists x D(x) \equiv \exists x C(x) \wedge \exists y D(y)$$

At the left-hand side of this semantic equivalence, we may choose *different* elements for x in $C(x)$ and in $D(x)$.

Likewise,
$$\forall x (\exists x C(x) \wedge D(x)) \equiv \forall x (\exists y C(y) \wedge D(x))$$

Some semantic equivalences

Important semantic equivalences in predicate logic:

$$\neg \exists x \phi \equiv \forall x \neg \phi$$

$$\neg \forall x \phi \equiv \exists x \neg \phi$$

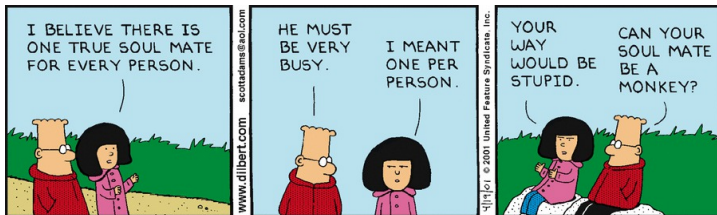
$$\exists x (\phi \vee \psi) \equiv \exists x \phi \vee \exists x \psi$$

$$\forall x (\phi \wedge \psi) \equiv \forall x \phi \wedge \forall x \psi$$

$$\exists x \exists y \phi \equiv \exists y \exists x \phi$$

$$\forall x \forall y \phi \equiv \forall y \forall x \phi$$

Dilbert's confusion



What Tina intends to say is

$$\forall x \text{ (} \textit{Person}(x) \rightarrow \exists y \text{ (} \textit{Person}(y) \wedge \textit{Soulmate}(x, y) \text{))}$$

But Dilbert interprets "there is one true soul mate for every person" as

$$\exists y \forall x \text{ (} \textit{Person}(x) \rightarrow \textit{Soulmate}(x, y) \text{))}$$

So Dilbert thinks Tina means all persons share the *same* soul mate y .

And he wonders what is the *range* of y . Does it include monkeys?

Some non-equivalences

The following are **not** semantic equivalences:

$$\exists x (C(x) \wedge D(x)) \not\equiv \exists x C(x) \wedge \exists x D(x)$$

$$\forall x (C(x) \vee D(x)) \not\equiv \forall x C(x) \vee \forall x D(x)$$

$$\forall x \exists y K(x, y) \not\equiv \exists y \forall x K(x, y)$$



Question

In each case,

- ▶ say which formula semantically entails the other one, and
- ▶ give a model in which one of the formulas holds and the other does not.

Some non-equivalences

$$A = \{p, q\} \quad C^{\mathcal{M}} = \{p\} \quad D^{\mathcal{M}} = \{q\}$$

$$\exists x (C(x) \wedge D(x)) \not\equiv \exists x C(x) \wedge \exists x D(x)$$

The left-hand side semantically entails the right-hand side, but not vice versa.

$$\mathcal{M} \not\models \exists x (C(x) \wedge D(x)) \quad \mathcal{M} \models \exists x C(x) \wedge \exists x D(x)$$

$$\forall x (C(x) \vee D(x)) \not\equiv \forall x C(x) \vee \forall x D(x)$$

The right-hand side semantically entails the left-hand side, but not vice versa.

$$\mathcal{M} \models \forall x (C(x) \vee D(x)) \quad \mathcal{M} \not\models \forall x C(x) \vee \forall x D(x)$$

Some non-equivalences

$$\forall x \exists y K(x, y) \not\equiv \exists y \forall x K(x, y)$$

The right-hand side semantically entails the left-hand side, but not vice versa.

$$A = \{p, q\} \quad K^{\mathcal{M}} = \{(p, q), (q, p)\}$$

$$\mathcal{M} \models \forall x \exists y K(x, y)$$

$$\mathcal{M} \not\models \exists y \forall x K(x, y)$$

Predicate logic

- ▶ syntax: constants and predicates
- ▶ universal and existential quantification
- ▶ semantics on the basis of models
- ▶ typical semantic equivalences and non-equivalences